

PREFACE

The following report has been prepared in accordance with the terms of Contract No. DA-36-034-ORD-1023 and constitutes the Final Report called for under the terms of that contract. It is divided into two parts (I and II) covering the salient engineering work and the mathematical investigations for the period July 1, 1952 through June 30, 1953. The actual carrying out of the calculations indicated in Part II was done under the terms of the contract. The mathematical preparations, i.e. the numerical analysis, programming and coding were carried out under the terms of Contract No. N-7-ONR-388, T. O. I and Contract No. N-6-ORI-139, T. O. I between the Institute for Advanced Study and the Office of Naval Research. Since the objectives of the three contracts are substantially overlapping it was felt desirable to include all this material in one report. In this fashion it is hoped to give the maximum possible information to all interested agencies.

John von Neumann
Project Director

Institute for Advanced Study

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PART I - ENGINEERING

I. INTRODUCTION

This report describes the operation of and engineering improvements on the electronic computer at the Institute for Advanced Study during the period from 1 July 1952 to 30 June 1953. The engineering discussion is restricted to those features of the present machine which were added during this period. The underlying design philosophy and a full technical description of the machine prior to this period is given in the final report on Contract Nos. W-36-034-ORD-7481 and DA-36-034-ORD-19 (Project No. TB3-0007 F).

APPENDIX

Arithmetic Test Code:

The arithmetic test code is intended to test for correct execution of the major machine orders. It is essentially a series of sub-routines, one for each order. Within each sub-routine standard operands are used by the order concerned and the result checked against a stored correct answer. In almost every case, two sets of operands are used: one to present maximum digit loads to the gate drivers concerned and one to present minimum loads. From a gate loading standpoint all numbers which may be handled by the order in question lie between these extremes.

If a given subroutine elicits an error, a card (12 words) is punched out identifying the type of error and the order concerned as well as all quantities entering into the computation. Note that an operand can be correctly stored in the memory but transferred incorrectly into the arithmetic unit. Therefore the operands are first positioned in the arithmetic unit (e.g. the multiplier is transferred from the memory into R_2 in the multiplication cases) then are temporarily stored from this position (by the two orders $R_2 \rightarrow R_1$, store for R_2 quantities such as the multiplier). It is these temporarily stored images of the operands which are provided at the error punch out.

The sequence in which processes are tested involves some important considerations. The machine can detect its own errors only by using processes which may themselves be in error. The comparison of the computed value with the stored correct answer necessarily involves a subtraction; and the discrimination on the zero or non-zero error must be made by a

conditional transfer of control. Therefore the transfer control orders are checked first. If these are correct, it is then possible to check the summation orders. If both are correct, then the other orders can be tested.

The scope of this test is summarized in the following table:

Arithmetic Test Sequence

<u>Test</u>	<u>Remarks</u>
1. Total Sum of Memory = 0	Memory contents check.
2. Transfer control	Tests all cases of conditional and unconditional transfers of control.
3. Summation (Digit)	Considering any stage of the adder, tests the 8 combinations of the two input digits and the carry.
4. Summation (Sign)	Tests the complement gate selector in the 8 combinations of the sign of the incoming number and the magnitude and +/- digits of the order.
5. R_2 to R_1	Tests $R_2 \rightarrow R^3$ gating under minimum and maximum digit loads. ($R^3 \rightarrow R_1$ transfer has already been tested by 4).
6. 1 Shift Right 2 Shift Left 4 Shift Right 8 Shift Left 16 Shift Right 32 Shift Left	Tests individual digits of the shift counting channel.
7. 31 Shift Right	Maximum digit load on shift count.
8. Gate loads	Tests all RI and RII gates under minimum and maximum digit loads.
9. Multiplication	Tests 8 combinations of sign of multiplier, sign of multiplicand, and Roundoff-No Roundoff.

- | | |
|--------------------|---|
| 10. Multiplication | Tests special case of small negative product which rounds to zero. |
| 11. Multiplication | Alternating Accept-Reject (i.e. multiplier alternates zeros and ones). Supplements 9. |
| 12. Division | Tests 4 combinations of sign of numerator and sign of denominator. |

Read Around Test Code:

This code tests the read around, or spill, properties of each point in the memory by clearing the four neighbors (left, right, upper, lower) of the point to zero K times and then storing a one at the point n times, after which the four neighbors are examined to see if any have been changed to ones. If any errors are found, a card is punched out giving the frequency n, the bombarding word (initially all ones), the stage and neighbor failing, and the other parameters of the code.

The code is stored initially in the upper half of the memory and first tests all points in the lower half at the specified initial frequency. Then the code transfers itself to the lower half and tests the upper half at the same frequency. After this the code is re-transferred to the upper half, the frequency increased by a specified increment, and the process repeated until the given maximum test frequency has been used.

When a given stage (i.e. memory cathode ray tube) has failed h times it is removed from the test, as much as possible, by changing the corresponding digit in the bombarding word to a zero. Another parameter in the code determines by what integer the last bombarded address is

increased to obtain the next test point. If this is other than one, then the necessary several passes are made through the half memory before testing is transferred to the other half.